

2) Six healthy subjects volunteered for the following study. Each subject carried out three experimental trials, (with a two hour break between trials) as follows:

Treatment A: 30 minutes upright seated taking a statistics exam

Treatment B: 15 minutes of moderate speed walking on a treadmill

Treatment C: 15 minutes bicycling at a moderate speed

The treatment order was randomized for each subject. The response variable was a measure of blood flow. A higher score meant a greater blood flow. The data for the six subjects is as follows:

Subject	Treatment A	Treatment B	Treatment C
1	52	29	39
2	33	32	25
3	49	38	48
4	62	45	43
5	40	38	37
6	46	34	30

- a) Carry out a test at the 0.05 level of significance of the Null Hypothesis of no difference between average blood flow for the three treatments, against the alternative that there is some difference.
- b) Explain to a non-statistician the interpretation of the conclusions from your analysis..